

1. EXECUTIVE SUMMARY

This report documents the successful completion of a 1-month virtual internship in **Data Science** at **Prodigy InfoTech**. The internship period spanned from **August 1, 2026, to August 31, 2026**. Over the course of this program, practical tasks involving data analysis, exploratory data visualization, predictive modeling, and machine learning pipeline development were executed, earning completion with outstanding remarks.

2. GENERAL INFORMATION

Candidate Name	Utkarsh Shukla
Domain / Field	Data Science
Organization	Prodigy InfoTech
Certificate ID / CIN	PIT/JUL25/05398
Internship Period	1 st August, 2026 – 31 st August, 2026 (1 Month)
Date of Certificate Issue	September 7, 2026
Official Contact	contact@prodigyinfotech.dev
Official Website	www.prodigyinfotech.dev

3. SCOPE OF WORK & KEY RESPONSIBILITIES

Exploratory Data Analysis (EDA) & Data Preprocessing

- Handled missing data, outliers, and data quality issues across diverse tabular datasets.
- Performed feature scaling, categorical variable encoding, and multi-variable correlation analyses.
- Evaluated demographic and statistical distributions to prepare raw data for machine learning modeling.

Data Visualization & Insights Generation

- Built clear, insightful visualizations (bar charts, distribution plots, correlation heatmaps) to interpret complex data patterns.
- Extracted actionable business and technical insights to inform feature engineering and modeling approaches.

Predictive Modeling & Model Evaluation

- Trained classification and regression models using supervised learning algorithms (e.g., Decision Trees, Random Forests, Linear/Logistic Models).
- Evaluated performance using metrics including Confusion Matrix, Precision, Recall, F1-Score, and Root Mean Squared Error (RMSE).

Code Quality & Documentation

- Maintained clean, well-documented Jupyter Notebooks and structured GitHub project repositories adhering to industry standards.

4. KEY PERFORMANCE INDICATORS & OUTCOMES

Program Completion Status	COMPLETED
Performance Remarks	Outstanding Remarks
Credential Verification	Verified via official QR code tracking system embedded on completion certificate.

5. CONCLUSION

The 1-month Data Science internship at Prodigy InfoTech provided practical hands-on experience in end-to-end data processing, exploratory visualization, and practical machine learning implementation. All technical tasks and project milestones were successfully accomplished with distinction.